

Predicting Stock Price Movements with News Sentiment: A Data Science Journey

A Comprehensive Analysis of Financial News, Technical Indicators, and Sentiment Correlation

Enhanced for Week 12: Production-Grade Portfolio Piece for Finance Sector

Week 12 Enhancements

This project has been transformed from a Week 1 analysis into a production-grade portfolio piece demonstrating:

Enhancement	Description	Impact
**Code Refactor	Modular `src/`	Maintainability
**Unit Testing*	35 pytest tests	Reliability +++
**CI/CD Pipelin	GitHub Actions	Quality Assuran
**Interactive D	Streamlit app f	Stakeholder Eng
**Documentation	Professional RE	Accessibility +

Quick Start

```
# Install and run the dashboard
pip install -r requirements.txt
streamlit run app/dashboard.py
```

Run Tests

```
pytest tests/ -v # 35 tests, all passing
```

Introduction: Decoding Market Sentiment

Can we predict stock market movements by analyzing the sentiment of financial news? This project explores that question through a comprehensive analysis of 1.4 million financial news headlines and six major tech stocks. Over Week 1, we built a complete analytical framework covering exploratory data analysis, quantitative technical analysis, and sentiment correlation-with some surprising discoveries along the way.

The Dataset: Foundation for Analysis

Our analysis began with the Financial News and Stock Price Integration Dataset (FNSPID), containing 1,407,328 financial news headlines from 2011-2020. Each record included:

- Headline: News text for sentiment analysis

Task 1: Exploratory Data Analysis - Uncovering Patterns

Methodology

We developed a modular analysis framework with 7 specialized notebooks, reusable utility functions, and comprehensive data quality checks. This foundation enabled systematic exploration across multiple dimensions.

Key Finding #1: Headline Characteristics

Analysis: Headlines average 73 characters (11 words), with a median of 64 characters. However, significant variation exists:

- Range: 3 to 512 characters

Key Finding #2: Publisher Distribution

Financial news follows a classic long-tail distribution: few publishers dominate, while many contribute occasionally.

Implications:

- Top publishers account for disproportionate article share

Key Finding #3: Temporal Publication Patterns

Temporal analysis revealed clear patterns:

- Peak days: Tuesday-Thursday most active

Key Finding #4: Topic Modeling

Using Latent Dirichlet Allocation (LDA), we identified 10 dominant financial news topics:

1. Earnings Reports

Data Quality Challenges

We encountered significant data quality issues:

- 96.02% invalid dates - Required robust parsing pipelines

Task 2: Quantitative Analysis - Technical Indicators

Moving Averages: Trend Identification

We calculated Simple Moving Averages (SMA) for 20, 50, and 200-day periods, plus Exponential Moving Averages (EMA) for 12 and 26-day periods across all six stocks.

Key Insights:

- SMA crossovers signal trend changes

RSI: Momentum Analysis

The Relative Strength Index (RSI) identifies overbought (>70) and oversold (<30) conditions.

Findings:

- NVDA shows highest RSI volatility, frequently entering extreme zones

MACD: Trend-Following Signals

Moving Average Convergence Divergence (MACD) identifies trend changes through:

- Bullish crossovers: MACD line crosses above signal (buy signal)

Financial Metrics: Risk-Return Profiles

We calculated comprehensive financial metrics for each stock:

Stock	Mean Daily Retu	Volatility	Sharpe Ratio	Max Drawdown
NVDA	0.19%	2.89%	Highest	-18.76%
AAPL	0.13%	1.80%	Moderate	-12.86%
AMZN	0.13%	2.18%	Moderate	-14.05%
META	0.11%	2.53%	Lower	-26.39%
MSFT	0.10%	1.69%	Moderate	-14.74%
GOOG	0.09%	1.73%	Lower	-11.10%

Key Insights:

- NVDA offers highest returns with highest volatility

Task 3: Correlation Analysis - Sentiment and Returns

Sentiment Analysis Methodology

We analyzed 55,230 headlines aligned to trading days using two complementary tools:

1. TextBlob: General-purpose sentiment polarity (-1 to 1)

Correlation Analysis: Critical Discovery

When correlating daily sentiment with daily stock returns, we encountered a critical limitation:

The Problem:

- Only 0.05-0.13% of trading days had matching news sentiment data

Root Cause Analysis

The date alignment issue stemmed from:

- News data: 2011-2020 (only 3.98% valid dates)

Interpretation

While we successfully:

- [DONE] Performed sentiment analysis on 55,000+ headlines

Key Findings and Insights

What We Learned

1. Pattern Recognition: Financial news follows predictable patterns in headlines, timing, and topics that could inform analysis.

Challenges Overcome

1. Date Parsing: Built robust pipelines for 96% invalid dates

Recommendations for Future Work

1. Improve Date Alignment: Obtain news data with better date coverage matching trading days
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Conclusion: Lessons from the Journey

This Week-1 project revealed that predicting stock movements from news sentiment is more complex than initially anticipated. While we successfully:

[DONE] Analyzed 1.4 million headlines

[DONE] Calculated comprehensive technical indicators

[DONE] Performed sentiment analysis on 55,000+ articles

[DONE] Built a robust analytical framework

We discovered that data quality and alignment are critical for meaningful correlation analysis. The lack of statistically significant correlations doesn't invalidate the approach-it highlights the need for better data alignment and alternative analytical methods.

The Bigger Picture

This project demonstrates real-world data science challenges:

- Data quality often trumps sophisticated algorithms

Foundation for Future Work

The framework we've built-comprehensive EDA, technical analysis, and sentiment analysis pipelines-provides a solid foundation. With improved data alignment and alternative approaches, we may yet uncover relationships between news sentiment and stock movements.

Technical Appendix

Tools and Libraries

- Data Processing: pandas, numpy

Key Metrics

- Headlines Analyzed: 1,407,328

Author: Data Science Team

Project: Week 1 - Stock Challenge (Enhanced Week 12)

Date: February 2026

Status: All tasks completed with production-grade enhancements

Week 12 Transformation Summary

From Analysis to Production

This project demonstrates the journey from initial analysis to a production-ready portfolio piece:

Phase	Week 1	Week 12
Code Structure	Notebooks + uti	Modular `src/`
Type Safety	None	Full type hints
Testing	None	35 unit tests
CI/CD	Basic	Multi-Python ma
Documentation	Basic README	Professional wi
Visualization	Static notebook	Interactive Str

Finance Sector Readiness

This enhanced project demonstrates key competencies valued in finance:

1. Reliability: Comprehensive testing ensures code correctness

Next Steps for Production

1. Deploy dashboard to cloud platform (Streamlit Cloud, AWS, etc.)
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This report represents a comprehensive analysis of financial news sentiment and stock price movements, transformed into a production-grade portfolio piece demonstrating reliability, professionalism, and business impact valued by finance sector employers.